Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/11/2010
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jimq@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
Organization Name:	University of Maryland		
PI Address 1:	Fire Protection Engineering		
PI Address 2:	3104G Jm Patterson		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Sunderland, Peter (University of Maryland)		
Grant/Contract No.:	NNX10AD98G		
Performance Goal No.:			
Performance Goal Text:			
	A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station. The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to		

Task Description:	use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. The gas flow rate will be adjusted such that the heat flux to the burner surface matches the heat required to vaporize the appropriate flow rate of liquid or solid fuel. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The flight objectives for BRE are: 1. Burn gaseous fuels in BRE to simulate the burning rates of condensed fuels in a quiescent microgravity environment. Instantaneous flame heat flux will be measured and used to control the rate of fuel supply. Fuels to be simulated include paper, plastic, and alcohols. 2. Burn gaseous fuels in BRE to estimate the ignition and extinction behavior of these condensed fuels in microgravity. 3. Use BRE to estimate the effects of burning at enhanced oxygen conditions in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	This project commenced on January 1, 2010. An experimental apparatus and wick burners were developed. These allow for observations of alcohol burning on porous wicks. Diagnostics include color imaging and gravimetric burning rate measurements. An extensive suite of tests have been performed for alcohols burning on porous wicks. The measurements correlate with Rayleigh number. This work is being written up for journal submission. A presentation on this project was given at NASA GRC on June 2, 2010. Michael J. Bustamante joined this project for his M.S. thesis work.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Articles in Peer-reviewed Journals	Dotson KT, Bustamante MJ, Sunderland PB, Quintiere JG. "Alcohol Burning Rates for Wicks at Various Orientations." Fire Safety Journal, in preparation, November 2010. Expected publication January 2011. , Nov-2010
Dissertations and Theses	Bustamante MJ. "Burning Rate Emulator." M.S. Thesis, University of Maryland, Department of Fire Protection Engineering, in preparation. November 2010. , Nov-2010
Papers from Meeting Proceedings	Dotson KT, Bustamante MJ, Sunderland PB, Quintiere JG. "Alcohol Burning Rates for Wicks at Various Orientations." Seventh Joint U.S. National Combustion Meeting. Atlanta, Georgia, March 2011. Seventh Joint U.S. National Combustion Meeting. Atlanta, Georgia, to be held March 2011. In preparation, November 2010. , Nov-2010